

FINAL PROJECT Dept. of CSE, BUET

2D Positional-Audio Sandbox



Team ABC – CSE 220

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Project at a Glance

A real-time **2D positional-audio engine**.

Place a listener and sources on a grid, paint walls, and hear – live – how distance, obstruction, material, and reflection change the sound:

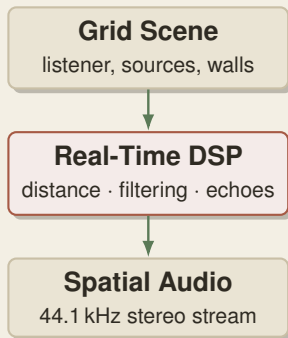
- **Muffling** as paths get obstructed
- **Panning** as sources move left / right
- Delayed, directional **echoes**
- Sound **bleeding through** walls

PYTHON

PYGAME

NUMPY

SOUNDDEVICE



Live, interactive · all in Python

Motivation: Beyond Empty Space

Free Space is Trivial

- Moving away $\rightarrow$ volume drops (gain $\propto 1/d$)
- Moving sideways $\rightarrow$ stereo pan (left / right)

In a void, spatial audio is just basic amplitude scaling.

Real Rooms Have Geometry

- **Corners & edges** – diffract and muffle highs
- **Solid walls** – absorb, let muffled sound bleed through
- **Surfaces** – generate delayed, directional echoes

The Central Challenge

“Can we simulate walls, materials, and echoes convincingly, live, using foundational signal-processing principles?”

IN SCOPE Discrete signals, convolution, recursive IIR, delays

REAL-TIME Causal stateful processing at 44.1 kHz

Goal: a real-time (44.1 kHz) interactive sandbox, built from stateful IIR filtering, delay lines, and 2D grid search.

Audio Input Pipeline: Three Operational Modes

MODE 1: AUDIO FILE

Format: .wav, .flac, .mp3

Pipeline: Eagerly loaded into float32 NumPy memory

Role: Instant testing with no audio hardware needed

MODE 2: LOCAL MIC

Format: Live microphone hardware

Pipeline: sounddevice circular stream buffer

Role: Real-time voice testing directly in the room

MODE 3: NETWORK UDP

Format: Remote Wi-Fi/Ethernet clients

Pipeline: UDP port 50007 to thread-safe queues

Role: Multi-source relay with client auto-reclaim

HARDWARE VERIFIED

Live Multi-Device Validation: Mode 3 verified across physical laptops on an active LAN, firewall traversal included.

Networked Streaming & Sampling Parameters

Network Plumbing (network_input.py)

- **Transport:** UDP port 50007 – drops packets over buffer stalls
- **Packet format:** 32-bit float PCM, 1024 samples per datagram
- **Lifecycle:** background listener → queues; idle timeout ≈ 5 s

Sampling & Latency Budget

- **Nyquist–Shannon:** $f_{\max} \approx 20$ kHz
 $\Rightarrow f_s = 44.1$ kHz (no aliasing)
- **Block size** $N = 1024$:
 $\Delta t = N/f_s \approx 23.2$ ms per buffer
- Short enough to feel instant, long enough for stable FFTs



Challenge 1: The Intuitive Starting Point

“Let’s start with the absolute basics: if a sound source is farther away, it should sound quieter. If it moves to your left, you should hear it in your left ear.”

THE CHALLENGE

How do we translate simple 2D grid coordinates (x, y) into natural volume attenuation and stereo balance — without complex 3D physics?

- **Distance:** amplitude must decay smoothly – no sudden dropoffs, no infinite volume at zero range
- **Direction:** horizontal displacement must pan smoothly without losing volume when centered

Challenge 1: Distance & Direction — Formulas

INVERSE-DISTANCE Distance attenuation

$$g(d) = \frac{1.0}{1.0 + k \cdot d}$$

$k = 0.08$ – tuned by ear on a 40×24 grid.

STEREO PAN Linear azimuthal panning

$$\text{pan} = \text{clip}\left(\frac{\Delta x}{20.0}, -1, 1\right), \quad \Delta x = x_{\text{src}} - x_{\text{lis}}$$

$$g_L = \frac{1 - \text{pan}}{2}, \quad g_R = \frac{1 + \text{pan}}{2}$$

COMBINED STEREO OUTPUT Per-sample signal vector

$$\mathbf{y}[n] = \begin{bmatrix} g_L g(d) x[n] \\ g_R g(d) x[n] \end{bmatrix}$$

Key Properties

- Constant total gain: $g_L + g_R = 1$ – no volume dip at center
- Smooth decay: no singularity at $d=0$
- Pure NumPy: `mono * gain * pan`

Challenge 1: Signal Scaling in Practice

SYLLABUS TIE-IN: SCALAR MULTIPLICATION

The fundamental operation:

$$y[n] = A \cdot x[n]$$

- Time-domain scalar multiplication – the simplest linear transform
- Spatial audio starts as amplitude modulation across two channels
- This is the *baseline* before any frequency-domain or delay-based effects

IMPLEMENTATION

Implementation Details

Pure NumPy vectorization: `mono * gain * pan`.
No per-sample Python loops – entire 1024-sample block processed in one NumPy broadcast.

Live Demo Cue

DEMO CUT Move a single source circularly around the stationary listener on an open grid – volume and stereo balance shift live.

Challenge 2: Obstacles & Acoustic Character

“A wall shouldn’t just be an ON/OFF mute switch. In real life, when someone speaks behind a door, you still hear them — it just sounds duller, muffled, like the treble was stripped away.”

THE CHALLENGE

How do we simulate acoustic “muffling” dynamically in real time — changing the spectral character of the signal, rather than just cutting its volume?

- Sound doesn’t just stop; high frequencies are absorbed far faster than low frequencies
- We need a lowpass filter whose cutoff adapts dynamically to obstacles

Filtering: Architecture & Artifact Discovery

INITIAL ATTEMPT: PER-BLOCK FFT

- **Intuition:** multiply by smooth cutoff in frequency domain via `rfft` / `irfft`
- Process independent 1024-sample blocks with raised-cosine taper
- **Catastrophic failure:** circular convolution without overlap-add wraps energy into block edges
- Produces a step discontinuity every 1024 samples $\Rightarrow$ **audible 43.06 Hz periodic buzzing artifact!**

SOLUTION: STATEFUL BUTTERWORTH IIR

- Replaced block FFT with a causal 2nd-order Butterworth lowpass
- Persistent initial condition states ($\mathbf{z}_i$) carried across blocks
- **Bit-continuous seams:** zero step discontinuities, zero boundary buzz

Engineering Lesson

Block-based FFT without overlap-add is non-causal circular convolution. Real-time audio requires causal, state-preserving recursive filtering.

Second-Order Butterworth IIR Lowpass

RECURSIVE FILTER

DIFFERENCE EQUATION

Second-order causal IIR transfer function:

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Time-domain difference equation:

$$y[n] = \sum_{k=0}^2 b_k x[n-k] - \sum_{k=1}^2 a_k y[n-k]$$

Second-Order Sections (SOS):

- Cascaded biquad representation for maximal numerical stability
- Cached filter coefficients keyed to (f_c, f_s)

ACOUSTIC CHARACTERISTICS

Why

Butterworth?

- **Maximally flat passband:** no ripple in audible spectrum before cutoff
- **Smooth roll-off:** -12 dB/octave attenuation strips high-frequency treble naturally
- **Dynamic cutoff:** $f_c \in [300, 4000]$ Hz dynamically adapts as geometry changes

IIR

SOS FORMAT

-12 dB/OCT

Stateful Filtering: Eliminating Boundary Glitches

BLOCK DELAY STATE (ZI) Streaming audio in 1024-sample chunks resets filters without state:

- **Direct Form II Transposed:** 2 delay registers $\mathbf{z} = [z_1, z_2]^T$ per channel
- **State carry-over across blocks:**
 $(y_B, \mathbf{z}_{\text{out}}) = \text{sosfilt}(\text{sos}, x_B, \mathbf{z}_i = \mathbf{z}_{\text{in}})$
- C^0/C^1 **continuity:** smooth sample transitions at block boundaries

STATEFUL ZI

C0 CONT.

O(N) CAUSAL

ENGINEERING HIGHLIGHTS

- $O(N)$ computation per block (faster than FFT)
- 100% artifact-free: zero 43 Hz boundary buzz
- Independent filter state for direct, transmission, and echo paths

Live Demo Cue

DEMO CUT Toggle a wall between source and listener – smooth, artifact-free roll-off live.

Challenge 3: Quantifying Obstruction

“A thin barrier directly in front of you shouldn’t sound identical to a sound that has to wind through a maze of three hallways to reach your ears.”

THE CHALLENGE

Sound waves diffract and bend around corners. How do we quantify “how obstructed” a path is in arbitrary grid geometry, and map that directly to our lowpass filter?

- Sound finds the shortest open path around barriers – not just the Euclidean line
- Every direction change (corner diffraction) strips additional high-frequency energy

Acoustic Diffraction via 8-Directional Pathfinding

YEN'S + A* SEARCH Routing on the 2D

grid (`pathfinding.py`)

- **8-directional:** costs 1.0 (ortho) / $\sqrt{2}$ (diag) – approximates Euclidean acoustics
- **Yen's k -shortest loopless paths** with A* (octile heuristic) for the primary acoustic path
- **Corner counting:** recorded whenever step direction $\Delta \mathbf{p}$ changes

Algorithmic technique applied directly to acoustic modeling.

CORNER-DRIVEN LOWPASS Mapping

geometry to DSP (`dsp_engine.py`)

$$f_c = \max(f_{\min}, f_{\text{base}} - \text{corners} \cdot \Delta f_{\text{corner}})$$

- $f_{\text{base}} = 4000$ Hz (open / 0 corners)
- $\Delta f_{\text{corner}} = 900$ Hz per corner
- $f_{\min} = 300$ Hz (prevents silence)

Distance gain uses routed path length:

$$g(d_{\text{path}}) = \frac{1.0}{1.0 + 0.08 \cdot d_{\text{path}}}$$

Engineering Insight: The “Phantom Corner” Bug

SUBTLE BUG: PHANTOM CORNERS

- **Symptom:** empty-grid diagonal paths sounded muffled
- **Root cause:** discrete raster steps along non-45° diagonals stair-step between ortho/diag moves
- Pathfinder counted each stair-step as a direction change $\implies$ 3–5 phantom corners in open space
- Caught by ear; confirmed by logging path vectors

THE FIX: LoS BYPASS

- Bresenham ray-cast:
`has_line_of_sight(src, lis, walls)`
- **Rule:** if straight line is clear, bypass corners entirely: $f_c = f_{\text{base}} = 4000 \text{ Hz}$
- Crisp highs in open space; corner muffling only behind real walls

Live Demo Cue

DEMO CUT Compare two sources at equal distance: clear LoS vs. behind a wall corner. Listen for the brightness drop.

Challenge 4: Sound Takes Time to Travel

“If sound bounces off a distant wall, it shouldn’t just be quieter. It travels a longer physical path, so it has to arrive later — like a real echo.”

THE CHALLENGE

Sound has a finite propagation speed. How do we model physical travel time in discrete-time digital audio without bogging down real-time performance?

- Instantaneous attenuation alone sounds unnatural and flat
- Reflected waves must arrive later, proportional to their extra travel distance

⇒ Next: translating physical distance into discrete-time sample shifts using circular buffers.

Discrete-Time Delay: The Operator

DISCRETE DELAY $x[n - D]$ **Signals tie-in:**
the delay operator

$$y[n] = x[n - D]$$

Continuous → **discrete mapping:**

$$\tau = \frac{\Delta d}{c_{\text{sound}}}, \quad D = \text{round}(\Delta d \cdot S_{\text{unit}})$$

Engine parameters ($f_s = 44.1$ kHz):

- $S_{\text{unit}} = 40$ samples/grid-unit (≈ 0.907 ms)
- $D_{\text{max}} = 8000$ samples (≈ 181 ms cap)
- Natural early reflections at room scale

SYLLABUS TIE-IN **The time-shift property**

- $x[n - D]$ is one of the fundamental discrete-time operations
- Combined with scaling $A \cdot x[n - D]$, it models both attenuation *and* propagation delay
- Foundation for echo modeling in Challenges 6–7

40 SAMPLES/UNIT

$f_s = 44.1$ kHz

Circular Ring Buffer Implementation

PERSISTENT CIRCULAR BUFFER Audio

arrives in chunks ($N=1024$) – storing full history is impossible.

Fixed-size circular NumPy array in `filter_state`, persistent write head w_{pos} :

$$\text{write_idx} = (w_{\text{pos}} + n) \bmod N_{\text{buf}}$$

$$\text{read_idx} = (w_{\text{pos}} + n - D) \bmod N_{\text{buf}}$$

Zero allocation per block $\implies$ predictable $\mathcal{O}(N)$.

ENGINEERING PROPERTIES

- Pre-allocated at source creation – no runtime `malloc`
- Modular arithmetic handles wraparound seamlessly
- Same buffer structure reused for all 4 echo slots (Challenge 6)

Live Demo Cue

DEMO CUT Place a source near a reflective wall; trigger a clap — hear the direct sound followed by a distinct, delayed secondary arrival.

Challenge 5: Total Occlusion & The Snapping Problem

“What happens when you close the last door and completely box a sound in? It shouldn’t suddenly snap to dead silence. Real walls let sound bleed straight through.”

THE CHALLENGE

When a room is fully sealed and no open path exists, how do we model acoustic penetration continuously — without jarring volume jumps or artificial floor hacks?

- **Wave penetration:** Acoustic energy travels directly through solid wall matter
- **Boundary continuity:** Sealing the final gap must fade smoothly without volume snaps

⇒ Next: why our initial “floor gain” shortcut failed, and how true material transmission replaced it.

Total Occlusion: Flat Floor vs. Physical Transmission

LEGACY FLAW: THE VOLUME SNAP

- **The shortcut:** 0 open paths $\rightarrow$ flat constant floor:
 $\text{gain}_{\text{floor}} \approx 0.02$ (-34 dB)
- **Perceptual failure:** closing the final gap violently snapped from routed audio to -34 dB. Unphysical and jarring.

FAMILY B: DIRECT TRANSMISSION

- **A distinct arrival:** energy traveling the straight Bresenham line through wall cells
- **Physical complement:**
 $\tau_{\text{transmission}} = 1.0 - \alpha_{\text{reflectivity}}$
- **Always active:** when sealed, routed audio drops to 0 and transmission takes over smoothly

The Architectural Fix

"No open path" no longer snaps to a floor. Only transmission contributes, yielding a perfectly continuous transition.

Through-Wall Transmission: Mathematics & Materials

MULTIPLICATIVE ATTENUATION Bresenham

line walk traces all crossed wall cells.

Compound transmission gain:

$$g_B = \frac{1.0}{1.0 + 0.08 \cdot d_{\text{Euclid}}} \prod_{w \in \text{crossed}} (1.0 - \alpha_w)$$

- Concrete ($\alpha=0.90$) $\implies$ transmits 10%
- Curtain ($\alpha=0.25$) $\implies$ transmits 75%
- Two curtains: $(0.75)^2 \approx 56\%$ – 5× more than one concrete wall

STEEPER LOWPASS CUTOFF

Solid barriers absorb highs more aggressively than air diffraction:

$$f_{C,B} = \max(300, 1500 - 500 \cdot N_{\text{walls}}) \text{ Hz}$$

Zero extra delay: straight geometric line
 $\implies$ no delay line needed.

THROUGH-WALL

NO SNAP

PRODUCT $(1-\alpha)$

Live Demo Cue

DEMO CUT

Box in a source – sound bleeds through, dark and muffled. Swap concrete for curtains to hear transmission brighten live.

Challenge 6: One Echo Isn't a Room

“A real room doesn't produce just one echo. If you clap between two walls, you hear sound bouncing off the left wall, the right wall, and the back corner — each arriving at a different time, from a different direction.”

THE CHALLENGE

Our early model tracked only a single secondary path ($k=2$). How do we discover, rank, and simulate multiple physically distinct reflections in real time?

- We want geometrically grounded early reflections off visible surfaces — not a generic reverb blur
- Each reflecting wall must sound like an independent source with its own delay, muffle, and angle

Family C: 12-Ray Radial Candidate Search

12-RAY RADIAL FAN The radial ray fan

(pathfinding.py)

- **360° sweep:** 12 rays at 30° increments from the source
- **Outward march:** steps out up to 20-cell radius until hitting a solid wall
- **Explicable:** every echo maps to a visible wall on the grid – no opaque reverb

VISIBILITY & HALF-SPACE FILTER A wall cell

qualifies as a true reflector only if:

- Source → Wall and Wall → Listener are unobstructed
- **Surface normal check:**
 $(\mathbf{x}_{\text{lis}} - \mathbf{x}_{\text{wall}}) \cdot \hat{\mathbf{n}}_{\text{wall}} > 0$
- Eliminates phantom echoes through the rear of occluding walls

Predicted loudness:

$$L(c) = \frac{0.75 \cdot \alpha_{\text{material}}}{1.0 + 0.05 \cdot 2 d_{\text{wall}}}$$

Top-4 selection: ranked by $L(c)$; best $M \leq 4$ sent to DSP.

Multi-Echo Delay Network

PARALLEL DELAY NETWORK **Four**

persistent ring buffers (`dsp_engine.py`)

- `echo_delay_buf_0...buf_3` in `filter_state`
- **Round-trip delay:**
 $D_i = \text{round}(2 d_{\text{wall},i} \cdot 110)$ samples
- ≈ 2.5 ms per grid unit $\implies$ 80–120 ms clear slapback
- **Echo gain:** $g_{C,i} = \frac{0.75 \cdot \alpha_i}{1.0 + 0.05 \cdot 2 d_{\text{wall},i}}$

MATERIAL-DEPENDENT CUTOFF Each echo

gets its own lowpass based on the reflecting wall's material:

$$f_{C,i} = \max(300, 3500 - (1 - \alpha_i) \cdot 3000) \text{ Hz}$$

- Hard wall ($\alpha = 0.90$): bright, crisp echo
- Soft curtain ($\alpha = 0.25$): heavily damped echo
- Each echo is independently IIR-filtered, delayed, and panned

Echo Spatial Panning & Demo

PER-WALL AZIMUTHAL PANNING Each echo pans toward its reflecting wall, not an averaged direction:

$$\Delta x_i = x_{\text{wall},i} - x_{\text{listener}}$$

Syllabus tie-in — multi-tap delay filter:

$$y[n] = \sum_{i=1}^M g_i \cdot x[n - D_i]$$

Each tap corresponds to a physically distinct reflecting surface.

4 INDEPENDENT ECHOES Every echo is a fully independent arrival:

- Its own delay (round-trip distance)
- Its own gain (distance + material)
- Its own lowpass (material cutoff)
- Its own stereo pan (wall direction)

Live Demo Cue

DEMO CUT Place a source between a concrete wall (left) and a wood wall (right) — hear two distinct, spatially separated echoes.

Challenge 7: Bringing It All Together

“We now have a primary diffracted arrival, through-wall transmission bleed, and up to four independent echoes. How do all these separate physical paths combine into a single stereo signal?”

THE CHALLENGE

How do we merge disparate acoustic phenomena into one coherent output — and what prevents the combined signal from distorting when multiple loud sources mix?

- Do we need a complicated blending heuristic to glue these arrivals together?
- When multiple sources sum, how do we guarantee signal stability?

The Culmination: Linear Superposition

SUMMED ACOUSTIC FAMILIES The complete
three-family model (`dsp_engine.py`)

$$\mathbf{y}_{\text{total}}[n] = \mathbf{y}_A[n] + \mathbf{y}_B[n] + \sum_{i=1}^M \mathbf{y}_{C,i}[n]$$

- $\mathbf{y}_A$ – routed path (corner lowpass + distance + LoS bypass)
- $\mathbf{y}_B$ – straight-line transmission (barrier lowpass + material product)
- $\mathbf{y}_{C,i}$ – up to 4 echoes (per-wall delay + muffle + pan)

No global re-normalization — inverse-distance laws already scale each contribution.

SUPERPOSITION PRINCIPLE The defining
law of linear systems:

$$\mathcal{T}\{x_1 + x_2\} = \mathcal{T}\{x_1\} + \mathcal{T}\{x_2\}$$

- Each path is an independent linear transform
- They combine by simple addition
- Signals & Systems foundation: complex propagation as parallel linear subsystems

Safety Limiting & The Full Engine Soundscape

CONDITIONAL TANH SOFT-CLIP Engineering

safety net (`audio_engine.py`)

- **Risk:** multiple sources sum $> \pm 1.0 \implies$ harsh clipping
- **Soft limiter:** $\mathbf{y}_{\text{mix}} = \tanh(\mathbf{y}_{\text{mix}})$ if $N_{\text{active}} > 1$
- Single source $\implies$ strictly linear
- Multiple sources $\implies$ smooth saturation

THE FULL ACOUSTIC SANDBOX All

phenomena running live, simultaneously:

- Direct diffracted path bending through doorways
- Muffled transmission bleeding through back wall
- Crisp slapback echoes off adjacent partitions

The Grand Demo Cue

DEMO CUT Activate complex multi-wall scene — hear diffraction, bleed, and echoes live at 44.1 kHz.

Challenge 8: Material Acoustic Diversity

“Should a heavy velvet curtain and a solid concrete bunker behave the same way? Obviously not — one absorbs treble and lets sound pass through; the other bounces sound like an acoustic mirror.”

THE CHALLENGE

How do we support diverse wall materials without introducing dozens of arbitrary, disconnected tuning knobs?

- Moving beyond binary “wall vs. empty grid” geometry
- A unified parameter where reflectivity and transmission loss are inherently coupled

Multi-Material Wall System: Hard, Medium, Soft

PALETTE SWATCHES Walls painted via sidebar swatches (`shared_state.py`):

- **Hard** (concrete): $\alpha_r=0.90$ – crisp echo, 0.10 transmission
- **Medium** (wood, default): $\alpha_r=0.60$ – balanced (0.40 transmission)
- **Soft** (curtain): $\alpha_r=0.25$ – porous (0.75 transmission)

UNIFIED ACOUSTIC COUPLING α_r

simultaneously controls three behaviors:

1. Echo gain: $g_{\text{echo}} \propto \alpha_r$
2. Echo cutoff:
 $f_c = \max(300, 3500 - (1 - \alpha_r) \cdot 3000)$ Hz
3. Through-wall bleed: $\tau = (1 - \alpha_r)$

Energy conservation: a material that transmits cannot also produce a loud reflection.

CONCRETE 0.90

WOOD 0.60

CURTAIN 0.25

Live Demo Cue

DEMO CUT

Paint a wall Concrete — loud slap-back. Repaint Curtain — echo dulls, bleed-through brightens.

System Architecture: Thread Isolation

STRICT THREAD SEPARATION The two

real-time worlds (`shared_state.py`)

- **Main thread** (Pygame, 60 FPS): drag-and-drop, wall painting, grid rendering
- **PortAudio callback** (44.1 kHz, ≈ 23 ms budget): hard real-time – must never block or allocate

The bridge: `get_snapshot()` grabs a brief lock, deep-copies positions and walls, releases immediately.

DELIBERATE TRADE-OFF Why not share

the audio thread's geometry cache?

- Cache in `filter_state` is mutated lock-free inside the callback
- Sharing risks priority inversion / buffer underruns
- **Choice:** render thread re-evaluates the cheap geometry functions independently (< 1 ms)

Architecture Takeaway

Tiny duplicated CPU work in exchange for zero lock contention and complete thread safety.

Live Grid Visualization: Seeing the Acoustics

REAL-TIME OVERLAYS As users draw walls, the grid shows which phenomenon is active (`render_engine.py`):

- **SOLID TEAL** – primary routed path (Family A)
- **DASHED ORANGE** – through-wall transmission (Family B)
- **MAGENTA RAYS** – echo reflections (Family C) with bounce markers

AUDIO-VISUAL CORRESPONDENCE Why this matters for the demo:

- Without visual cues, spatial audio is invisible
- Place a wall → teal line bends, orange dashed line appears
- Box in a source → teal disappears, dashed remains, magenta rays bounce inside

Presentation Tip

During the demo, point to the colored path lines to explain which formula is feeding the speakers.

Visualizing the Spectrum: Real-Time FFT Panel

LOG-SPACED FFT BINS Sidebar spectrum

analyzer (`viz_engine.py`)

- $|X[k]| = |\text{rfft}(x[n])|$ drawn as animated bars
- **Log binning** (40 Hz $\rightarrow$ 22 kHz):
mimics pitch hearing
- **Focus:** 300–4000 Hz (wall filter range)

MAGNITUDE $|X[k]|$

LOG BINS

TIME vs. FREQUENCY DOMAIN

- **Time domain:** waveforms show distance and delay
- **Frequency domain:** spectrum exposes filtering directly
- **Visual proof:** audience sees HF bars drop as wall is added

Live Demo Cue

DEMO CUT Toggle a wall: watch HF bars drop instantly as sound muffles.

Summary: The Signal Processing Chain

FROM SIGNALS TO SOUNDSCAPES

1. **Distance & panning:** signal scaling, spatial balance
2. **Stateful IIR lowpass:** causal 2nd-order Butterworth filters
3. **Diffraction routing:** 8-dir Yen's A^* with corner penalty
4. **Discrete delay lines:** circular buffers for propagation
5. **Through-wall bleed:** continuous fading, no floor snap
6. **Multi-wall echoes:** 12-ray fan, top-4 reflectors
7. **Multi-material:** Hard / Medium / Soft coupling
8. **Superposition:** direct summation into stereo out

A complete from-scratch layered audio model, built from recursive IIR filtering, delay networks, and 2D grid search.

Summary: System Highlights

CAPABILITIES

- **Flexible input:** file sandbox, local mic, multi-device UDP relay (firewall-tested)
- **Dual real-time visualization:** grid path overlays + log-spaced FFT spectrum analyzer
- **Clean architecture:** strict thread isolation via `SharedState`; zero dead code

TESTING & QUALITY

- `test_dsp_engine.py` – 24 tests, 117 assertions
- `test_pathfinding.py` – 16 test blocks
- `test_viz_engine.py` & `test_udp_sender.py`
- Modular black-box design; no leftover dead code

3 INPUT MODES

3 WALL FAMILIES

117+ ASSERTIONS

ZERO DEAD CODE

Live Demonstration

THE SANDBOX IN ACTION

Interactive Demonstration

Fast scene exploration via **Hotkeys [1]–[8]** or sidebar dropdown.

1. **Presets [1]–[3] (Occlusion & Diffraction):** LoS baseline → single-wall shadow → multi-corner maze
2. **Presets [4]–[6] (Echoes & Stereo):** Slapback echo off concrete → through-wall bleed → ping-pong stereo
3. **Presets [7]–[8] (Materials & Multi-Source):** Curtain dampening vs. dual independent audio sources
4. **Dynamic Sandbox:** drag source/listener live, draw custom walls, toggle materials

HOTKEYS 1–8

8 PRESET SCENES

117+ PASSING TESTS

LIVE DEMO

Thank You



Questions & Live Sandbox Exploration